

Session 4: PAKE-0, Assumptions, and Channels

Slides prepared by UMBC Protocol Analysis Lab

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Passwords

Password:

Log In

[Forgot Your Password?](#)

Can help prove your identity to somebody.

We've used these forever.

Should be kept secret and hard to guess.

Often memorized. . .

Don't store these in plain-text anywhere.

PAKE-0

Alice

Bob

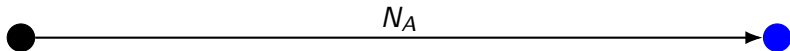
shared secret password: **pw**

● = Reception ● = Transmission

PAKE-0

Alice

Bob



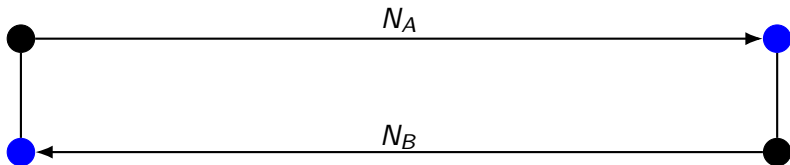
shared secret password: **pw**

● = Reception ● = Transmission

PAKE-0

Alice

Bob



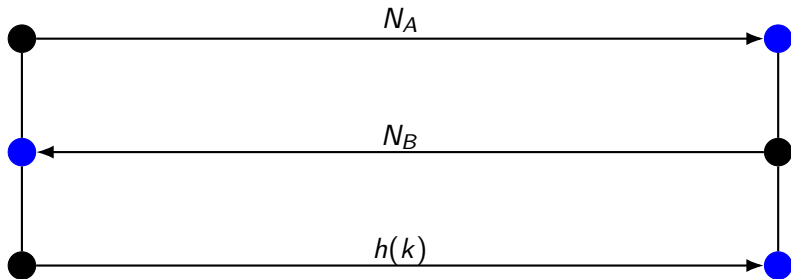
shared secret password: **pw**

● = Reception ● = Transmission

PAKE-0

Alice

Bob



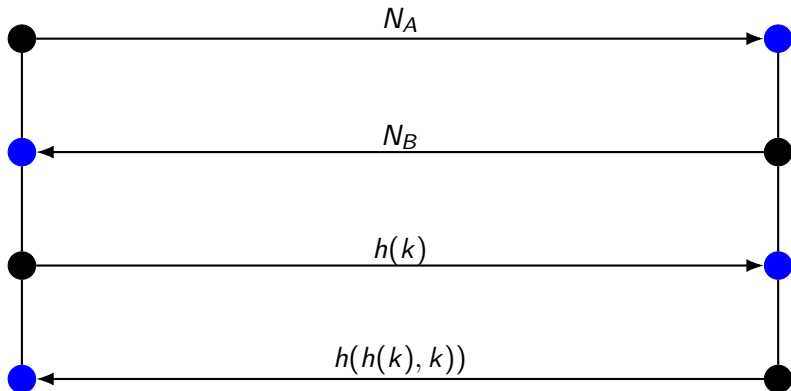
shared secret password: **pw** $k = h(pw, Alice, Bob, N_A, N_B)$

● = Reception ● = Transmission

PAKE-0

Alice

Bob



shared secret password: **pw** $k = h(pw, Alice, Bob, N_A, N_B)$

● = Reception ● = Transmission

PAKE-0 textbook diagram

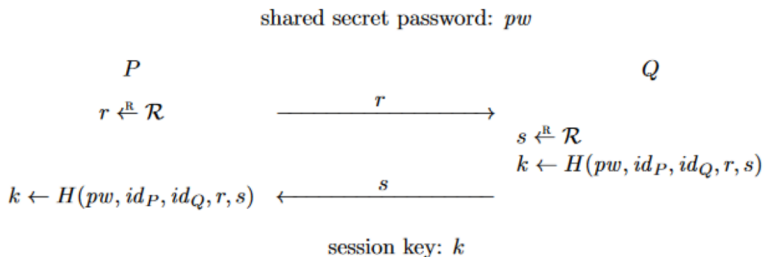


Figure 21.13: Protocol PAKE_0

Figure: Boneh, Dan, and Victor Shoup. "A graduate course in applied cryptography." Draft 0.5 (2020).

Assumptions in CPSA (1)

Data origination assumptions:

uniq-orig (e.g., nonce)

Definition (unique origination)

A value *uniquely originates* in a bundle if it originates on exactly one strand. Note that if the unique origination point of a value is on a regular strand, then it cannot be produced in a “create” instance because that would constitute a second point of origination.

Assumptions in CPSA (2)

non-orig (e.g., private key)

Definition (non-origination)

A value is *non-originating* in a bundle if it does not originate on any strands. A simple lemma can show that a value is non-originating only if it is not carried in any message in the bundle.

Assumptions in CPSA (3)

pen-non-orig (e.g., pre-shared password)

Definition (penetrator non-originating)

A value is *penetrator non-originating* in a bundle of a protocol if it does not originate on any penetrator strands. Clearly, such a value cannot be produced in a “create” instance.

Homework

Model more in CPSA

Model more in CPSA

Now that you've seen PAKE modeled, can you model a more challenging protocol?

Model your meeting protocol from Homework 1!

Survey

Fill in details about the survey here!